

Respuestas Trabajo Práctico 7

$$1. \quad (a) \int f(x)dx = \frac{x^5}{5} + \frac{x^3}{3} - 2x^2 + C \quad (c) \int f(x)dx = \frac{1}{4}(2x-1)e^{2x} + C$$

$$(b) \int f(x)dx = -\operatorname{sen} x + \operatorname{tg}(\pi x) + C$$

$$2. \quad (a) \frac{5}{3}x^3 + 2x - \frac{1}{2}x^{-2} + C$$

$$(c) e^x + \frac{7}{2}x^2 + C$$

$$(b) \frac{3}{5}x^{\frac{5}{3}} - \frac{2}{3}x^{\frac{3}{2}} + x + C$$

$$(d) e^{x-2} + 2\operatorname{sen} x - \frac{x^4}{4} + C$$

$$3. \quad (a) \frac{1}{6}(4x+1)^{\frac{3}{2}} + C$$

$$(d) -\frac{\cos(x^2+2x)}{2} + C$$

$$(g) \ln|\ln x| + C$$

$$(b) \frac{1}{3}\operatorname{sen}(3x) + C$$

$$(e) \frac{1}{2}\sqrt{x^4+2} + C$$

$$(h) \frac{\operatorname{sen}^2(x)}{2} + C$$

$$(c) -\frac{1}{5}e^{-5x} + C$$

$$(f) \frac{1}{2}e^{x^2} + C$$

$$4. \quad (a) \frac{1}{5}\operatorname{sen}(5x)(2x+1) + \frac{2}{25}\cos(5x) + C$$

$$(d) -\frac{\ln x}{x} - \frac{1}{x} + C$$

$$(b) \frac{x^3}{3}\ln x - \frac{x^3}{9} + C$$

$$(e) -x^2\cos x + 2x\operatorname{sen} x + 2\cos x + C$$

$$(c) \frac{x}{2}e^{2x} + \frac{3}{4}e^{2x} + C$$

$$(f) x(\ln x)^2 - 2x\ln x + 2x + C$$

$$5. \quad (a) \frac{e^{x^2}}{2}x^2 - \frac{e^{x^2}}{2} + C$$

$$(c) \frac{1}{x}\cos\left(\frac{1}{x}\right) - \operatorname{sen}\left(\frac{1}{x}\right) + C$$

$$(b) 2e^{\sqrt{x}}(\sqrt{x}-1) + C$$

$$(d) x\operatorname{arc\,tg} x - \frac{1}{2}\ln(x^2+1) + C$$

$$6. \quad (a) -e^{1/x} + c$$

$$(f) (1+x)\operatorname{arc\,tg}(\sqrt{x}) - \sqrt{x} + c$$

$$(b) \frac{1}{2}x^2\ln(x) - \frac{1}{4}x^2 + c$$

$$(g) -\frac{3}{4}(3-2x-x^2)^{\frac{2}{3}} + c$$

$$(c) \operatorname{sen}(\ln x) + c$$

$$(h) x\operatorname{senh}(x) - \cosh(x) + c$$

$$(d) \frac{1}{5}(x^2+1)^{\frac{5}{2}} - \frac{1}{3}(x^2+1)^{\frac{3}{2}} + c$$

$$(i) \frac{1}{2}\ln(x^2+1) + \operatorname{arc\,tg}(x) + c$$

$$(e) -\frac{1}{2(\ln x)^2} + c$$

$$7. \quad (b) \quad \text{i. } f'(x) = \sqrt{1+x^2}$$

$$\text{ii. } f'(x) = \operatorname{sen}(x^2)$$

$$\text{iii. } f'(x) = -\cos(e^x)$$

$$\text{iv. } f'(x) = 3x^2 \frac{\operatorname{sen}(x^3)}{x^3}$$

8. (a) es creciente en $(0; +\infty)$,

(b) es cóncava hacia arriba en $(0; +\infty)$.

10. (a) $\frac{147}{2}$

(c) $\frac{78728}{9}$

(e) 0

(g) $\frac{\pi}{4}$

(b) $\frac{e^4 - 1}{2e^4}$

(d) $-\ln 2 + \frac{1}{4}$

(f) $-\frac{4}{3}$

11. (a) $\frac{2\sqrt{2}}{3} - \frac{1}{3}$

(c) $2\ln 2 - \frac{3}{4}$

(b) $\frac{1}{7}$

(d) $2 - e$

12. (a) $\int_{-2}^2 \sqrt{2-x} \, dx = \frac{16}{3}$

(c) $\int_{-\pi}^0 -\sin x \, dx + \int_0^{\pi} \sin x \, dx = 4$

(b) $\int_0^2 -(x^2 - 2x) \, dx = \frac{4}{3}$

13. (a) 5

(b) 3

(c) $\frac{1}{\ln 2}$

14. (a) $c = \sqrt{50} - 3$

(b) $c = \sqrt[3]{5}$

15. (a) $2\ln|x+1| - \ln|x-4| + C$

(c) $\frac{1}{x+1} + 2\ln|x| - 2\ln|x+1| + C$

(b) $-2\ln|x+1| + \ln|x-2| + 2\ln|x+3| + C$

(d) $\frac{1}{5}\ln|x-2| - \frac{1}{10}\ln|x^2+1| - \frac{2}{5}\arctg x + C$

16. (a) $-\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C$

(c) $\frac{\operatorname{tg}^3 x}{3} + \frac{\operatorname{tg}^5 x}{5} + C$

(b) $\frac{\sin^3 x}{3} - \frac{2\sin^5 x}{5} + \frac{\sin^7 x}{7} + C$

(d) $-\frac{\sin(4x)}{8} + \frac{\sin(2x)}{4} + C$